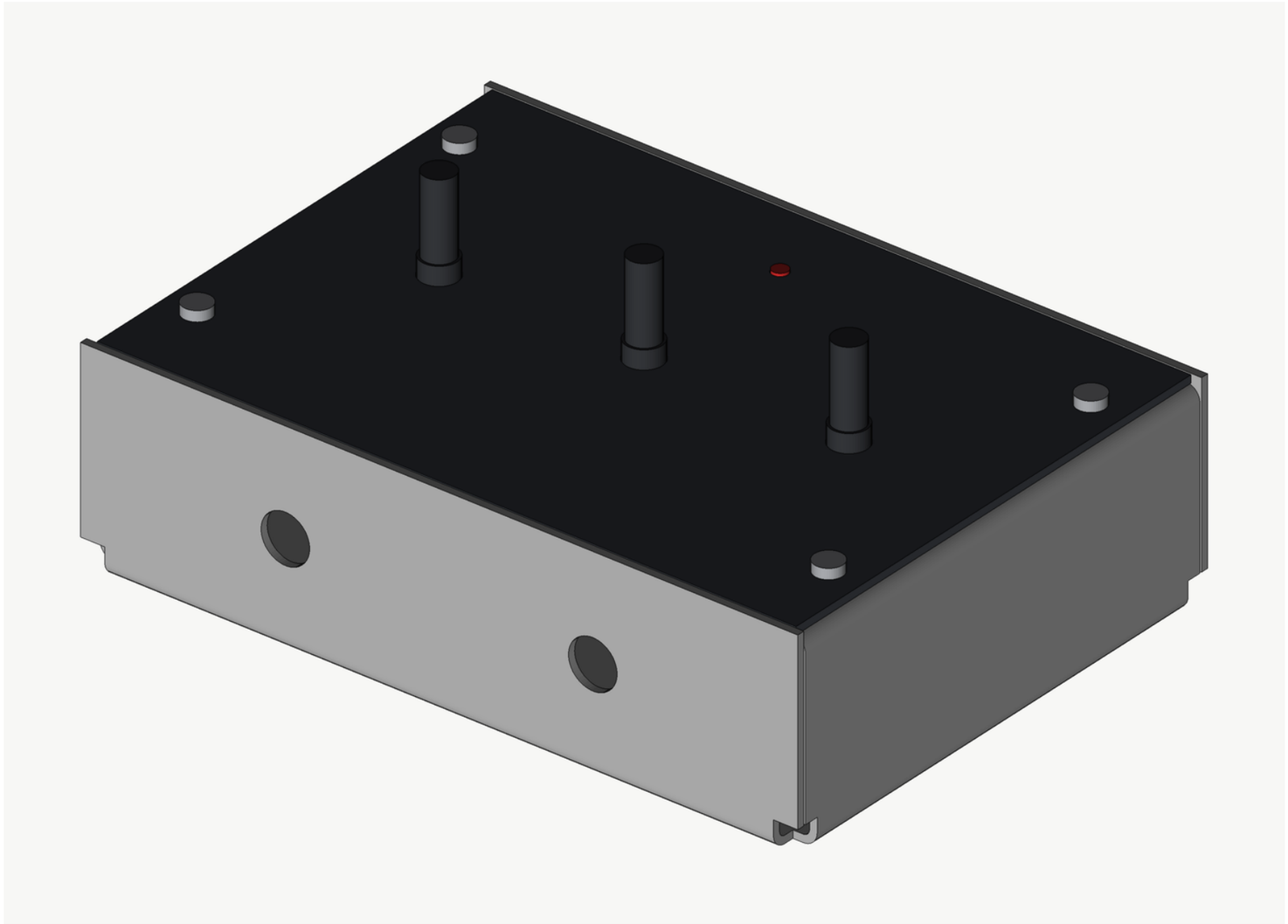


Checks: 35 pass, 0 warn, 0 fail. All 35 are on the last sheet.

Every check passes.

Sizes

Size	mm
Overall, everything	140 × 100 × 62.5
tray, formed	140 × 100 × 40
tray, flat	234.77 × 174.88
lid, formed	136.4 × 96.4 × 1.5
lid, flat	136.4 × 96.4

Parts list

#	Component	Qty	Source
1	tray	1	sheet metal
2	lid	1	sheet metal
3	screw_M3x6_pan	4	stand-in
4	rubber_foot_12x4	4	stand-in
5	pot_9mm	3	stand-in
6	led_3mm	1	stand-in
7	control_board_pcb	1	stand-in

Parameters

Parameter	Value
shop	jlccnc
material	Aluminum 5052
thickness	1.5
W	140
D	100
H	40
corner_relief	2
corner_gap	0.4
lid_clearance	0.3
lip	14
pot_body	10

NOTES

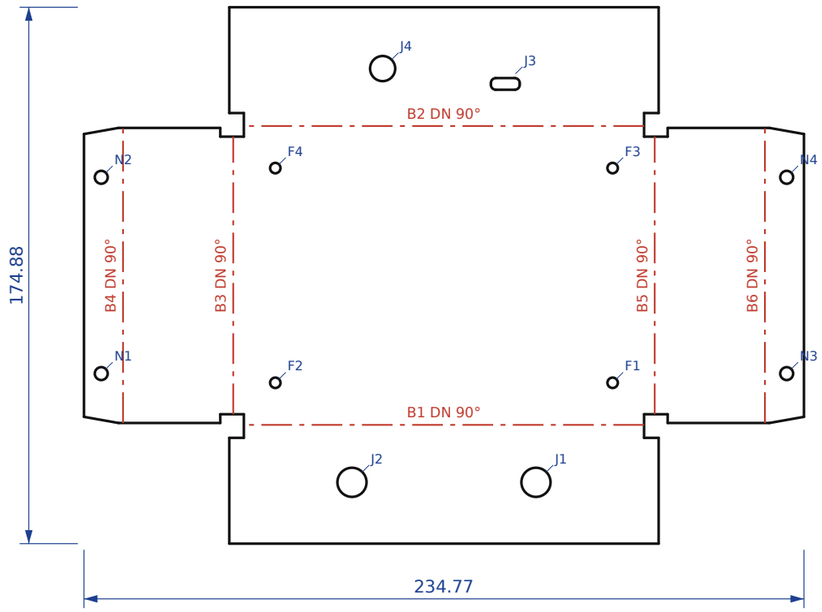
1. Inside bend radius R1.25, K-factor 0.415.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Inside radius 1.25 mm = published outer corner radius 2.75 - T.
5. K 0.415 is the middle of JLCCNC's design-guide range, not shop data. JLCCNC unfolds the STEP you send, so K only affects qfold's own flat pattern.
6. Actual sheet thickness 1.42 mm for nominal 1.5.

Pedal box: assembly

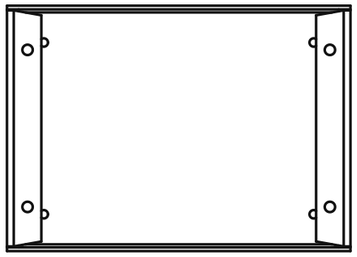
qfold

PART	SCALE	SHEET
ASSEMBLY	NTS	1 of 5
MATERIAL	UNITS	REV / DATE
JLCCNC Aluminum 5052, t=1.5	mm	A 2026-11-09

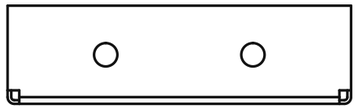
Formed size (outside) 140 × 100 × 40



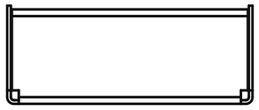
FLAT PATTERN 234.77 × 174.88



TOP (1:2.5)



FRONT (1:2.5)



RIGHT (1:2.5)

Holes (flat pattern, from lower-left)

Tag	Feature	X	Y	Size
F1	foot screw M3	172.38	52.44	Ø3.4
F2	foot screw M3	62.38	52.44	Ø3.4
F3	foot screw M3	172.38	122.44	Ø3.4
F4	foot screw M3	62.38	122.44	Ø3.4
J1	1/4" jack in	147.38	20	Ø9.5
J2	1/4" jack out	87.38	20	Ø9.5
J3	USB-C	137.38	149.88	9.4×3.8 R1.6
J4	DC barrel jack	97.38	154.88	Ø8.2
N1	PEM CLS-M3 nut	5.62	55.44	Ø4.22
N2	PEM CLS-M3 nut	5.62	119.44	Ø4.22
N3	PEM CLS-M3 nut	229.14	55.44	Ø4.22
N4	PEM CLS-M3 nut	229.14	119.44	Ø4.22

Bends

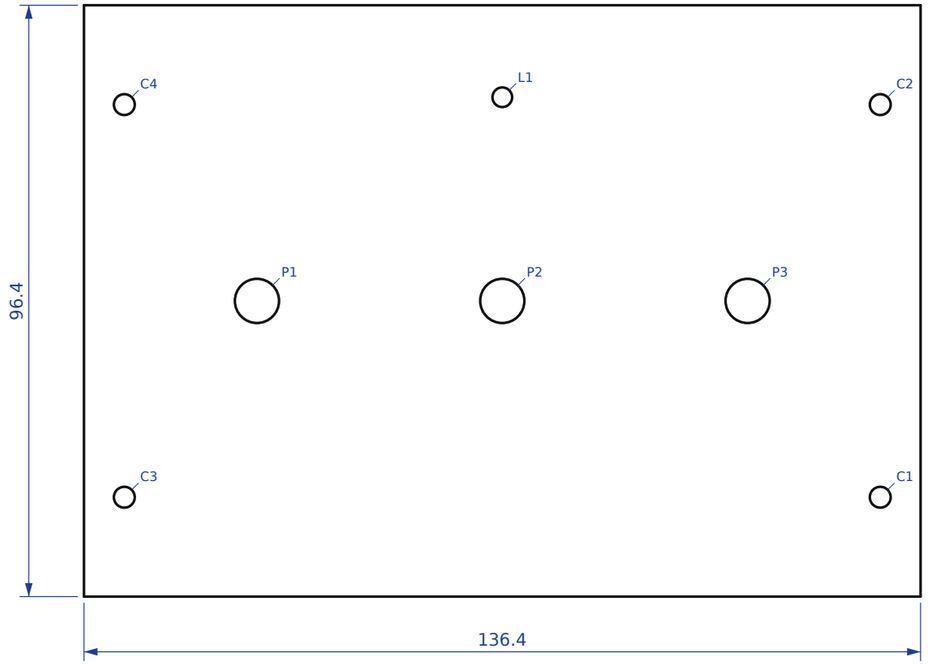
Tag	Bend	Dir	Angle	In. R	Length	Allow.	Centre line
B1	front	DN	90°	R1.25	130.5	2.94	Y 38.72
B2	back	DN	90°	R1.25	130.5	2.94	Y 136.16
B3	right	DN	90°	R1.25	90.5	2.94	X 48.66
B4	right_wall_lip	DN	90°	R1.25	96.2	2.94	X 12.72
B5	left	DN	90°	R1.25	90.5	2.94	X 186.1
B6	left_wall_lip	DN	90°	R1.25	96.2	2.94	X 222.04

NOTES

1. Inside bend radius R1.25, K-factor 0.415.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Inside radius 1.25 mm = published outer corner radius 2.75 - T.
5. K 0.415 is the middle of JLCCNC's design-guide range, not shop data. JLCCNC unfolds the STEP you send, so K only affects qfold's own flat pattern.
6. Actual sheet thickness 1.42 mm for nominal 1.5.
7. Flat pattern: outside face shown. DN = bend away from viewer. The DXF is the cutting master; the STEP is the formed part.

tray		qfold
PART **TRAY**	SCALE 1:2	SHEET 2 of 5
MATERIAL JLCCNC Aluminum 5052, t=1.5	UNITS mm	REV / DATE A 2026-11-09

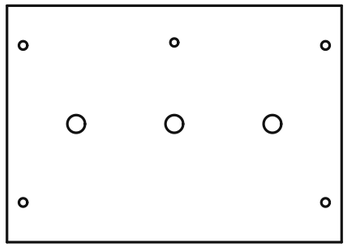
Formed size (outside) 136.4 × 96.4 × 1.5



FLAT PATTERN 136.4 × 96.4

Holes (flat pattern, from lower-left)

Tag	Feature	X	Y	Size
C1	M3 clearance	129.82	16.2	Ø3.4
C2	M3 clearance	129.82	80.2	Ø3.4
C3	M3 clearance	6.58	16.2	Ø3.4
C4	M3 clearance	6.58	80.2	Ø3.4
P1	pot (M7 bushing)	28.2	48.2	Ø7.2
P2	pot (M7 bushing)	68.2	48.2	Ø7.2
P3	pot (M7 bushing)	108.2	48.2	Ø7.2
L1	3 mm LED	68.2	81.4	Ø3.2



TOP (1:2.5)

FRONT (1:2.5)

RIGHT (1:2.5)

NOTES

1. Inside bend radius R1.25, K-factor 0.415.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Inside radius 1.25 mm = published outer corner radius 2.75 - T.
5. K 0.415 is the middle of JLCCNC's design-guide range, not shop data. JLCCNC unfolds the STEP you send, so K only affects qfold's own flat pattern.
6. Actual sheet thickness 1.42 mm for nominal 1.5.
7. Flat pattern: outside face shown. DN = bend away from viewer. The DXF is the cutting master; the STEP is the formed part.

lid		qfold
PART **LID**	SCALE 1:1	SHEET 3 of 5
MATERIAL JLCCNC Aluminum 5052, t=1.5	UNITS mm	REV / DATE A 2026-11-09

Part	Tag	Feature	Size	X	Y	Z	Face	Facing
tray	F1	foot screw M3	Ø3.4	15	15	0	base	−Z down
tray	F2	foot screw M3	Ø3.4	125	15	0	base	−Z down
tray	F3	foot screw M3	Ø3.4	15	85	0	base	−Z down
tray	F4	foot screw M3	Ø3.4	125	85	0	base	−Z down
tray	J1	1/4" jack in	Ø9.5	40	0	20	front_wall	−Y front
tray	J2	1/4" jack out	Ø9.5	100	0	20	front_wall	−Y front
tray	J3	USB-C	9.4×3.8 R1.6	50	100	15	back_wall	+Y back
tray	J4	DC barrel jack	Ø8.2	90	100	20	back_wall	+Y back
tray	N1	PEM CLS-M3 nut	Ø4.22	131.62	18	38.5	right_wall_lip	+Z up
tray	N2	PEM CLS-M3 nut	Ø4.22	131.62	82	38.5	right_wall_lip	+Z up
tray	N3	PEM CLS-M3 nut	Ø4.22	8.38	18	38.5	left_wall_lip	+Z up
tray	N4	PEM CLS-M3 nut	Ø4.22	8.38	82	38.5	left_wall_lip	+Z up
lid	C1	M3 clearance	Ø3.4	131.62	18	40	lid	+Z up
lid	C2	M3 clearance	Ø3.4	131.62	82	40	lid	+Z up
lid	C3	M3 clearance	Ø3.4	8.38	18	40	lid	+Z up
lid	C4	M3 clearance	Ø3.4	8.38	82	40	lid	+Z up
lid	P1	pot (M7 bushing)	Ø7.2	30	50	40	lid	+Z up
lid	P2	pot (M7 bushing)	Ø7.2	70	50	40	lid	+Z up
lid	P3	pot (M7 bushing)	Ø7.2	110	50	40	lid	+Z up
lid	L1	3 mm LED	Ø3.2	70	83.2	40	lid	+Z up

NOTES

1. Hole centres on the outside surface, in assembly coordinates.
2. Facing: the direction the hole opens to, out of the part.

Holes as assembled

qfold

PART	SCALE	SHEET
PEDAL BOX	NTS	4 of 5
MATERIAL	UNITS	REV / DATE
JLCCNC Aluminum 5052, t=1.5	mm	A 2026-11-09

Checks: 35 pass, 0 warn, 0 fail

Check	Result	
tray: valid solid	valid=True, solids=1, faces=77	PASS
tray: no self-intersection	folded fuse loses -7.28e-12 mm³, flat fuse loses -2.18e-11 mm³ (limit 5.0e-05)	PASS
tray: flat pattern matches fold	folded−flat volume 190.583148 mm³, bend allowance predicts 190.583148	PASS
tray: hole to bend	tightest 3.52 mm (right_wall_lip PEM CLS-M3 nut to bend right_wall_lip), need ≥ 3.00	PASS
tray: hole to edge	closest 3.51 mm (left_wall_lip PEM CLS-M3 nut), need ≥ 2.25	PASS
tray: web between holes	narrowest 31.50 mm (back_wall USB-C / DC barrel jack), need ≥ 1.00	PASS
tray: holes inside their face	every hole lies within its face	PASS
tray: minimum hole size	all holes ≥ 1.00 mm	PASS
tray: hardware edge distance	4 hardware holes meet their centre-to-edge minimum	PASS
tray: bend corner relief	bends end ≥ 2.83 mm apart, need ≥ 1.50	PASS
tray: flange length	shortest flat flange 11.25 mm (bend right_wall_lip), need ≥ 5.25	PASS
lid: valid solid	valid=True, solids=1, faces=14	PASS
lid: no self-intersection	folded fuse loses 0.00e+00 mm³, flat fuse loses 0.00e+00 mm³ (limit 1.9e-05)	PASS
lid: flat pattern matches fold	folded−flat volume 0.000000 mm³, bend allowance predicts 0.000000	PASS
lid: hole to edge	closest 4.88 mm (lid M3 clearance), need ≥ 2.25	PASS
lid: web between holes	narrowest 28.00 mm (lid pot (M7 bushing) / 3 mm LED), need ≥ 1.00	PASS
lid: holes inside their face	every hole lies within its face	PASS
lid: minimum hole size	all holes ≥ 1.00 mm	PASS
tray.step reads back	1 solid(s), volume 49773.024 mm³ (model 49773.024)	PASS
lid.step reads back	1 solid(s), volume 19473.683 mm³ (model 19473.683)	PASS
assembly.step reads back	15 solid(s), volume 87017.138 mm³ (model 87017.138)	PASS
lid rests on the lips	1 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
screw heads on the lid	4 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
pots against the lid	3 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
pots on the PCB	3 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
LED in its hole	1 clear of it, no overlap	PASS
feet on the base	4 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
pcb clear of tray	overlap 0.000000 mm³, closest 8.559 mm, need ≥ 2	PASS
faces of one part keep their gaps	15 unjoined face pairs, tightest 0.400 mm (tray: front_wall vs right_wall), need ≥ 0.2	PASS
no part overlaps another	15 top-level parts, none overlapping	PASS
screws through lid and nut	4 parts, worst 0.000 mm across, 0.000 mm along, 0.000°	PASS
tray: fits sheet	flat 234.8 × 174.9 mm, sheet limit 800 × 800	PASS
tray_flat.dxf reads back	11 circles, 36 lines, 4 arcs, 6 bends, cut 1077.7 mm, extent 234.765 × 174.883	PASS
lid: fits sheet	flat 136.4 × 96.4 mm, sheet limit 800 × 800	PASS
lid_flat.dxf reads back	8 circles, 4 lines, 0 arcs, 0 bends, cut 586.2 mm, extent 136.400 × 96.400	PASS

NOTES

1. Run on every build; also in report.json.
2. Limits come from the project's rules (a shop's, or qfold's defaults in dfm.py).

Checks		qfold
PART **PEDAL BOX**	SCALE NTS	SHEET 5 of 5
MATERIAL JLCCNC Aluminum 5052, t=1.5	UNITS mm	REV / DATE A 2026-11-09